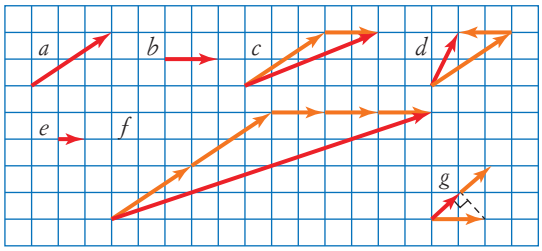
YEAR 11 SPECIALIST MATHS INVESTIGATIVE TASK 1 SOLUTIONS

Vectors

1 The answers are in green. One square = 1 unit × 1 unit.



2 **a** $5\mathbf{i} - 3\mathbf{j}$ **b** $-\mathbf{i} + 5\mathbf{j}$ **c** $12\mathbf{i} - 16\mathbf{j}$

d $\sqrt{3^2 + (-4)^2} = 5$ **e** $\frac{3}{5}\mathbf{i} - \frac{4}{5}\mathbf{j}$ or $0.6\mathbf{i} - 0.8\mathbf{j}$ **f** $-5\mathbf{i} + 15\mathbf{j}$

g 3 **[1 mark each]**

h $\cos(\theta) = \dfrac{\mathbf{a}\cdot\mathbf{c}}{|\mathbf{a}||\mathbf{c}|} = \dfrac{3}{\sqrt{5} \times \sqrt{2}} \approx 0.9486... \Rightarrow \theta \approx 18°$ **[2]**

i $p = \mathbf{a}\cdot\hat{\mathbf{b}} = 2 \times \frac{3}{5} + 1 \times \frac{4}{5} = \frac{2}{5}$ (or 0.4) **[2]**

j Parallel: $(\mathbf{a}\cdot\hat{\mathbf{c}})\hat{\mathbf{c}} = \dfrac{\mathbf{a}\cdot\mathbf{c}}{|\mathbf{c}|}\hat{\mathbf{c}} = \dfrac{3}{\sqrt{2}} \times \dfrac{\mathbf{i} + \mathbf{j}}{\sqrt{2}} = 1.5\mathbf{i} + 1.5\mathbf{j}$ **[2]**

Perpendicular: $\mathbf{a} - (1.5\mathbf{i} + 1.5\mathbf{j}) = 2\mathbf{i} + \mathbf{j} - 1.5\mathbf{i} - 1.5\mathbf{j} = 0.5\mathbf{i} - 0.5\mathbf{j}$ **[1]**

k $4\hat{\mathbf{c}} = \dfrac{4(\mathbf{i} + \mathbf{j})}{\sqrt{2}} = 2\sqrt{2}\mathbf{i} + 2\sqrt{2}\mathbf{j}$ **[2]**

l Let the vector be $x\mathbf{i} + y\mathbf{j}$. Then $(x\mathbf{i} + y\mathbf{j})\cdot(3\mathbf{i} - 4\mathbf{j}) = 0 \Rightarrow 3x - 4y = 0$ **[1]**

Magnitude is $10 \Rightarrow x^2 + y^2 = 100$. **[1]**

$3x - 4y = 0 \Rightarrow 4y = 3x \Rightarrow y = \dfrac{3x}{4}$ **[1]**

Substitute into $x^2 + y^2 = 100 \Rightarrow x^2 + \dfrac{9x^2}{16} = 100 \Rightarrow \dfrac{25x^2}{16} = 100$

$x^2 = 100 \times \dfrac{16}{25} = 64 \Rightarrow x = \pm 8$ **[2]**

$y = \dfrac{3x}{4} = \pm 6$ **[1]**

$8\mathbf{i} + 6\mathbf{j}$ or $-8\mathbf{i} - 6\mathbf{j}$ **[2]**

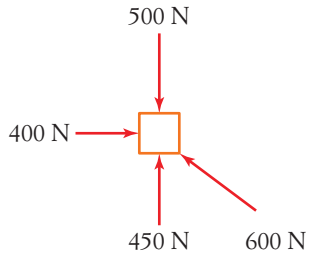
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3 a $\mathbf{CD} = \mathbf{CO} + \mathbf{OD} = -\mathbf{OC} + \mathbf{OD} = \mathbf{OD} - \mathbf{OC}$ **[2]**

b $\mathbf{OA} = \frac{1}{2}\mathbf{BA} = \frac{1}{2}\mathbf{CD} = \frac{1}{2}\mathbf{OD} - \frac{1}{2}\mathbf{OC}$ **[3]**

c $\mathbf{AD} = \mathbf{AO} + \mathbf{OD} = -\mathbf{OA} + \mathbf{OD} = -\frac{1}{2}\mathbf{OD} + \frac{1}{2}\mathbf{OC} + \mathbf{OD} = \frac{1}{2}\mathbf{OC} + \frac{1}{2}\mathbf{OD}$ **[4]**

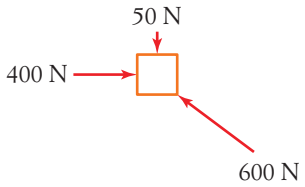
4 a



[2]

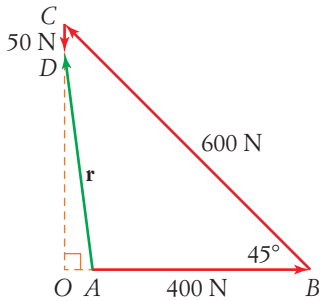
b A net force of $500 - 450 = 50$ N south. **[1]**

c Replace the 500 N to the south and the 450 N to the north with a single force of 50 N to the south.



[2]

d Draw a diagram for addition of vectors.



[3]

The green vector is the resultant **r**.

$OC = 600 \sin(45°) = \frac{600\sqrt{2}}{2} = 300\sqrt{2} \Rightarrow OD = 300\sqrt{2} - 50 \text{ N} (\approx 374 \text{ N})$ **[2]**

$OB = 600 \sin(45°) = \frac{600\sqrt{2}}{2} = 300\sqrt{2} \Rightarrow OA = 300\sqrt{2} - 400 \text{ N} (\approx 24 \text{ N})$ **[2]**

$|\mathbf{r}|^2 = (300\sqrt{2} - 50)^2 + (300\sqrt{2} - 400)^2 = 522\,500 - 270\,000\sqrt{2}$ (using CAS) **[1]**

$|\mathbf{r}| = \sqrt{522\,500 - 270\,000\sqrt{2}} \approx 375 \text{ N}$ **[1]**

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e The crate will (just) move. **[1]**

$\tan \angle OAD = \dfrac{OD}{OA} = \dfrac{300\sqrt{2} - 50}{300\sqrt{2} - 400} = 15.4246... \Rightarrow \angle OAD \approx 86.3°$ **[2]**

This is $(90 - 86.3)° = 3.7°$ west of north. **[1]**

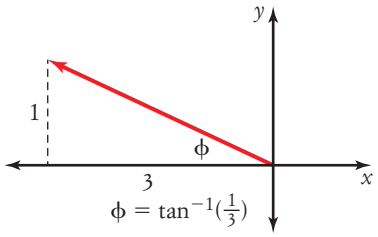
Answer: The crate will (just) move in a direction N3.7°E. **[1]**

5 a $d^2 = (-3)^2 + 1^2 = 10 \Rightarrow d = \sqrt{10}$ km ≈ 3.16 km **[2]**

$\tan(\theta) = \dfrac{1}{-3} = -\dfrac{1}{3} \Rightarrow \theta = \tan^{-1}\left(-\dfrac{1}{3}\right)$

$\theta \approx 162°$ so N72°W or 288°T **[3]**

Alternatively, students can use the diagram below.



$\theta = 18°$ so N72°W or 288°T **[3]**

b The resultant speed will be in the direction $-3\mathbf{i} + \mathbf{j}$. **[1]**

Superman's flight at 100 km/h + wind velocity = final velocity. **[1]**

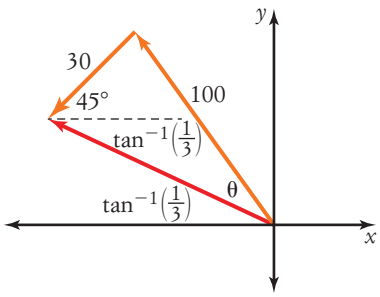
Draw a diagram.

The wind blows *from* the north-east. **[3]**

Use the sine rule.

$\dfrac{\sin(\theta)}{30} = \dfrac{\sin\left(45 + \tan^{-1}\left(\dfrac{1}{3}\right)\right)}{100}$

$\sin(\theta) = \dfrac{30 \sin\left(45 + \tan^{-1}\left(\dfrac{1}{3}\right)\right)}{100}$



$\sin(\theta) = 0.2683...$

$\theta = 15.5648... \approx 16°$ since $\theta < 90°$. (Keep number in calculator.) **[2]**

He must head 16° more northerly/clockwise from the criminal's direction.

He must fly at N(72° − 16°)W, i.e N56°W or (288° + 16°)T = 304°T **[2]**

c Use the cosine rule.

Let the top angle $= A = 180° - 45° - \tan^{-1}\left(\dfrac{1}{3}\right) - 15.5648... = 101.0002...°$

[1]

$a^2 = b^2 + c^2 - 2bc\cos(A)$

$a^2 = 900 + 10\,000 - 2\times30\times100\times\cos(101.0002....°)$

$a = 109.7491...$

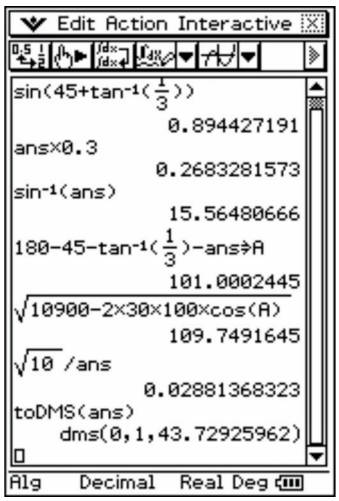
Superman's true speed = 109.7491... km/h. **[2]**

Time taken = distance ÷ speed = $\sqrt{10}$ ÷ 109.7491... = 0.02881... hours

[1]

Converting to DMS gives about 1 minute and 44 seconds. **[1]**

A sample screenshot for a student using a CAS calculator is given below.



d $(2\mathbf{i} + 5\mathbf{j}) - (-3\mathbf{i} + \mathbf{j}) = 5\mathbf{i} + 4\mathbf{j}$ **[1]**

e $|-3\mathbf{i} + \mathbf{j}| + |5\mathbf{i} + 4\mathbf{j}| + |2\mathbf{i} + 5\mathbf{j}| = \sqrt{10} + \sqrt{41} + \sqrt{29} \approx 14.95$ or nearly 15 km. **[2]**